**EXPERIMENT 13**

# Parse Tree Generation for Given Sentences using CFG

## AIM

To construct a Context-Free Grammar and generate structural parse trees for sentences using Python and NLTK.

## PROCEDURE

1. Formalize production rules describing grammatical phrases (Sentence, Noun Phrase, Verb Phrase, Prepositional Phrase).
2. Instantiate the CFG and NLTK ChartParser.
3. Pass tokenized sentences into the parser to construct tree hierarchies.
4. Display visual text representations of the parse trees.
5. Verify the structural constituency of each grammatical component.

## PROGRAM

```
import nltk
from nltk import CFG

grammar = CFG.fromstring("""
    S -> NP VP
    NP -> Det Adj N | Det N | 'Alice' | 'Bob'
    VP -> V NP | V NP PP | V
    PP -> P NP
    Det -> 'the' | 'a'
    Adj -> 'young' | 'intelligent'
    N -> 'student' | 'book' | 'library'
    V -> 'read' | 'borrowed'
    P -> 'in' | 'from'
""")

parser = nltk.ChartParser(grammar)
sentence = "Alice borrowed a book from the library".split()

print("Sentence:", " ".join(sentence))
print("=" * 60)

for tree in parser.parse(sentence):
    print("Tree String Representation:")
    print(tree)
    print("\nAscii Graphical Parse Tree:")
    tree.pretty_print()
```

## OUTPUT

```
> python3 experiment_13.py
```

```
Sentence: Alice borrowed a book from the library
============================================================
Tree String Representation:
(S
  (NP Alice)
  (VP
    (V borrowed)
    (NP (Det a) (N book))
    (PP (P from) (NP (Det the) (N library)))))

Ascii Graphical Parse Tree:
             S
  ___________|___________
 |                       VP
 |    ____________________|_________________
 |   |          NP                       PP
 |   |       ___|___                   ___|____
 NP  V     Det      N                 P        NP
 |   |      |       |                 |     ___|___
Alice borr  a      book              from  Det     N
     owed                                |      |
                                        the  library

[Process completed with exit code 0]
```

## RESULT

Thus, parse tree generation for natural sentences was successfully executed using Context-Free Grammar parsing.